Introduction to Typst

A modern, expressive, and blazing-fast typesetting system

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October 14, 2026

Abstract — This document serves as a clean, idiomatic example of a document authored in **Typst**. It demonstrates page setup, typography, mathematical formulas, formatted tables, code listings, callout admonitions, and cross-referencing in a dark mode (black and white) aesthetic.

1. Introduction

Typst is a markup-based typesetting system designed to be as powerful as LaTeX while offering a much simpler, more intuitive user experience. Unlike legacy tools, Typst compiles in milliseconds, offers helpful compile-time diagnostics, and features a clean syntax inspired by Markdown.

Key Advantages

- **Instant Compilation**: Incremental compilation provides real-time PDF feedback as you type.
- **Consistent Syntax**: Markup, math, and scripting share a cohesive, predictable design.
- **Modern Layout Engine**: Built-in support for flexible grids, borders, and modern typography.

Formatting text is straightforward:
- **Bold text** is written with `*bold*`
- *Italic text* is written with `_italic_`
- `Monospace text` is written with backticks
- Lists can be unordered (`- bullet`) or numbered (`+ item`):
  1. First ordered step
  2. Second ordered step
     - Nested bullet point
     - Additional detail
  3. Final conclusion step

2. Mathematical Typesetting

Math formulas in Typst are enclosed in dollar signs (`$`).

Inline equations sit comfortably within text: $E = mc^2$ or $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Block equations are created by inserting spaces or linebreaks inside `$ ... $`:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \tag{1}$$

We can easily cross-reference this Gaussian distribution via Equation 1.

Typst also handles matrices, derivatives, integrals, and limits with clean syntax:

$$M = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \tag{2}$$

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1, \quad \int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2} \tag{3}$$

3. Tables and Figures

Tables can be structured with `#table` and wrapped in `#figure` to provide automatic captions and numbering:

Aspect	Typst	LaTeX
Syntax	Intuitive, Markdown-like	Complex macro-heavy
Speed	Milliseconds	Multi-pass, slower
Error diagnostics	Precise with line snippets	Cryptic errors
Package ecosystem	Integrated packages (`@preview`)	CTAN / TeX Live

Table 1: Comparison between Typst and traditional LaTeX workflows.

As shown in Table 1, Typst simplifies modern publishing workflows without sacrificing output quality.

4. Code Listings

Syntax highlighting works cleanly on dark backgrounds:

```python
def compute_fibonacci(limit: int) -> list[int]:
    """Generate Fibonacci sequence up to limit items."""
    fib = [0, 1]
    while len(fib) < limit:
        fib.append(fib[-1] + fib[-2])
    return fib

print(compute_fibonacci(8))
```

5. Custom Admonition Component

Typst allows you to write functions that encapsulate reusable UI components:

> **Helpful Tip**

> Run `typst watch example.typ` in your terminal to enable automatic live preview in your favorite PDF reader.

6. Conclusion

With its lightweight syntax, instant compilation, and modern programmatic flexibility, Typst is an excellent choice for research papers, homework assignments, resumes, and technical documentation.